A Mathematically Rigorous Framework for Cross-Exchange Price Discovery

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DRAFT v1.2.0

Abstract

We develop a self-contained mathematical theory of price oracles for digital assets traded across n heterogeneous centralised exchanges.

(I) Consolidated Order Book. We embed n limit order books into a common rational price grid and define the COB map via a greedy arbitrage-exhaustion algorithm. The fair price f_t is justified as the *Chebyshev centre* of the post-exhaustion spread, the unique price minimising worst-case regret for both counterparties.

(II) DAN Convergence. True no-arbitrage is unachievable due to fees and latency; we define the ε-NA cone and prove the DAN drives the residual crossed spread into this cone exponentially fast at rate γ_k proportional to deployed capital.

(IV) Arbitrage-Driven Feed Detection. The DAN's bots are simultaneously economic actors and data reporters. Any falsified feed creates a phantom arbitrage whose execution reveals the manipulation through slippage, without cross-referencing other bots.

(VI) Manipulation Security and Optimal Delay. We formalise the reserve-constrained attack (book-emptying with early cancellation) and show it tightens the manipulation cost bound. We derive the optimal publication delay Δt^* balancing manipulation cost against staleness front-running (Theorem 4.10).

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1 Consolidated Order Book

1.1 Price Grids and Embedding

Definition 1.1. Let $\mathcal{E} = \{1, \dots, n\}$. Exchange $i \in \mathcal{E}$ has tick $\eta_i \in \mathbb{Q}_{>0}$ and grid $\mathcal{G}_i = \{k\eta_i : k \in \mathbb{N}_0,\ k \le K_i\}$. Write $\eta_i = p_i/q_i$ in lowest terms and set

$$D := \operatorname{lcm}(q_1, \dots, q_n), \qquad d_i := D\eta_i \in \mathbb{N}, \qquad \eta := \frac{\gcd(d_1, \dots, d_n)}{D} \in \mathbb{Q}_{>0}$$

The *finest common grid* is $\mathcal{G} := \{p_k := k\eta : 0 \le k \le K\}$ with $K := \max_i K_i$ and $m := |\mathcal{G}|$.

Lemma 1.2. *$\mathcal{G}_i \subseteq \mathcal{G}$ for every $i \in \mathcal{E}$.*

Proof. Write $k\eta_i = kd_i/D$. Since $g := \gcd(d_1, \dots, d_n)$ divides d_i, we have $kd_i/D = (kd_i/g) \cdot \eta$, an integer multiple of η in $[0, K\eta]$. □

Definition 1.3. The snapshot of exchange i at time t is $x_i^{(t)} = (b_i^{(t)}, a_i^{(t)}) \in \mathbb{R}_+^{2m}$, where $b_{i,k}^{(t)}$ (resp. $a_{i,k}^{(t)}$) is the resting bid (resp. ask) depth at price p_k on exchange i at time t. We set $b_{i,k}^{(t)} = a_{i,k}^{(t)} = 0$ for $p_k \notin \mathcal{G}_i$. The aggregate depth profiles are

$$B^{(t)} := \textstyle\sum_i b_i^{(t)} \in \mathbb{R}_+^m, \qquad A^{(t)} := \textstyle\sum_i a_i^{(t)} \in \mathbb{R}_+^m \tag{1}$$

We equip $\mathbb{R}_+^{2m}$ with the ℓ^1 norm $\|x\|_1 = \sum_k (|b_k| + |a_k|)$.

1.2 The COB Map and the Fair Price

Definition 1.4. Profiles (B, A) satisfy the *NA condition* if $\beta(B) := \max\{k : B_k > 0\} < \alpha(A) := \min\{k : A_k > 0\}$. The *residual spread* is the interval $[p_{\beta(B)}, p_{\alpha(A)}]$.

Definition 1.5. Given a NA pair (B, A) with residual spread $[p_\beta, p_\alpha]$, define

$$f(B, A) \ := \ \frac{p_\beta + p_\alpha}{2} \tag{2}$$

This is the *Chebyshev centre* of $[p_\beta, p_\alpha]$: the unique point minimising $\max(f - p_\beta,\ p_\alpha - f)$. It is the unique price at which a buyer and seller both receive non-negative surplus relative to their best available counterparty, and is characterised by the minimax-regret property:

$$f(B, A) \ = \ \operatorname*{arg\,min}_{p \in [p_\beta, p_\alpha]} \max\bigl(p - p_\beta,\ p_\alpha - p\bigr)$$

When the residual spread is one tick wide ($\alpha = \beta + 1$, the generic case in a liquid book), (2) reduces to the standard bid-ask midpoint. When the spread is wider, f is the unique price equidistant from both sides of the market, i.e. the price at which the exchange is equally uninformative about whether it should be bid or offered.

Remark 1.6. A natural alternative is the volume-weighted average price (VWAP) of trades executed during COB exhaustion: $f_{\text{vwap}} = \sum_k p_k v_k / \sum_k v_k$ where v_k is matched volume at p_k. When there is a large matched volume (deep overlap), f_{vwap} and f can differ significantly. We prefer (2) for two reasons: (i) it depends only on the *residual* spread, not on the path of matching, making it path-independent and hence manipulation-resistant to order-sequencing attacks; and (ii) when no overlap exists, f_{vwap} is undefined while f is always well-defined.

Theorem 1.7. *Algorithm 1 below terminates in at most $2m$ iterations. Its output $(\tilde{B}, \tilde{A})$ is the unique element of $\mathbb{R}_+^m \times \mathbb{R}_+^m$ satisfying: (i) the NA condition; (ii) $B - \tilde{B} = A - \tilde{A}$; (iii) $\tilde{B} \le B$ and $\tilde{A} \le A$ componentwise. The fair price f_t is a well-defined Borel-measurable function of $(x_i^{(t)})_{i \in \mathcal{E}}$.*

Input: $(B, A) \in \mathbb{R}_+^m \times \mathbb{R}_+^m$, $B, A \not\equiv 0$
Output: Residual NA pair $(\tilde{B}, \tilde{A})$ and fair price f_t
$(\tilde{B}, \tilde{A}) \leftarrow (B, A)$;
$b \leftarrow \beta(\tilde{B})$;
$a \leftarrow \alpha(\tilde{A})$
while $b \geq a$ **do**
  $v \leftarrow \min(\tilde{B}_b, \tilde{A}_a)$
  $\tilde{B}_b \leftarrow \tilde{B}_b - v$;
  $\tilde{A}_a \leftarrow \tilde{A}_a - v$
  if $\tilde{B}_b = 0$ **then** $b \leftarrow \max\{k < b : \tilde{B}_k > 0\}$
  if $\tilde{A}_a = 0$ **then** $a \leftarrow \min\{k > a : \tilde{A}_k > 0\}$
end
return $(\tilde{B}, \tilde{A})$;
$f_t \leftarrow (p_b + p_a)/2$

Algorithm 1: COB Arbitrage-Exhaustion Map

1.3 Exact Lipschitz Stability

Theorem 1.8. *For any two aggregate profile pairs (B, A) and (B', A'),*

$$|f - f'| \;\leq\; \frac{\eta}{2}\left(\left\|B - B'\right\|_1 + \left\|A - A'\right\|_1\right) \tag{3}$$

Proof. Write $\delta B := B' - B = \delta B^+ - \delta B^-$. Adding mass $\|\delta B^+\|_1$ to B can advance β by at most $\|\delta B^+\|_1 / \eta$ ticks (each tick level p_k requires at least η mass to create a new non-zero entry or to increase β past it). Removing $\|\delta B^-\|_1$ can retract β by at most $\|\delta B^-\|_1 / \eta$ ticks. Combined: $|\beta' - \beta| \leq \|\delta B\|_1 / \eta$, and symmetrically $|\alpha' - \alpha| \leq \|\delta A\|_1 / \eta$. Since $f = \eta(\beta + \alpha)/2$:

$$|f - f'| = \frac{\eta}{2}|(\beta' - \beta) + (\alpha' - \alpha)| \leq \frac{\eta}{2}(|\beta' - \beta| + |\alpha' - \alpha|) \leq \frac{1}{2}(\|\delta B\|_1 + \|\delta A\|_1)$$

□

Remark 1.9. The ℓ^1 norm of a depth perturbation equals the dollar volume of orders added or removed. Theorem 1.8 states that the oracle price moves by at most $\eta/2$ per dollar of volume perturbed: the precise quantitative link between market liquidity and oracle robustness.

2 DAN: Architecture and Convergence

2.1 The ε-NA Cone

Exact no-arbitrage ($\Delta_k = 0$) is unachievable: transaction fees set a minimum profitable spread $c_{j,k} > 0$, and latency prevents instantaneous reaction.

Definition 2.1. For cluster C_k with aggregate profiles $B^{C_k} := \sum_{i \in C_k} b_i^{(t)}$ and $A^{C_k} := \sum_{i \in C_k} a_i^{(t)}$, the *residual crossed spread* is

$$\Delta_k(t) \;:=\; \left[p_{\beta(B^{C_k})} - p_{\alpha(A^{C_k})}\right]^+$$

the amount by which the cluster best bid exceeds the cluster best ask after applying Algorithm 1. The *ε-NA cone* is $\mathcal{K}_\varepsilon := \{\Delta_k \le \varepsilon\}$.

Remark 2.2. The natural floor is $\mathcal{K}_{c_{j,k}}$: the fee-limited cone below which no bot executes. Capital beyond what is needed to reach this floor brings no additional accuracy improvement.

2.2 Architecture: Clusters, Bots, and Servers

Definition 2.3. Let $(\mathcal{F}, d)$ be a finite metric space of co-location facilities with $d : \mathcal{F} \times \mathcal{F} \to \mathbb{R}_+$ the median round-trip latency and $\ell(i, f) := d(\xi_i, f)$ the latency from facility f to exchange i. Fix $\tau_{\max} > 0$. A $(K, \tau_{\max})$*-cluster partition* is $(C_1, \ldots, C_K)$ with $\bigsqcup_k C_k = \mathcal{E}$ and $\operatorname{diam}(C_k) := \max_{i,j \in C_k} \ell(i, j) \le \tau_{\max}$ for all k.

A *placement* is a set $S \subseteq \mathcal{F}$ of bot-server locations. The *observation latency* of exchange i under S is $\lambda(i, S) := \min_{f \in S} \ell(i, f)$, ie. the fastest reporting facility of that exchange.

Definition 2.4. Bot $j \in [N_k]$ in cluster C_k holds capital $q_{j,i}^{(t)} \ge 0$ on exchange $i \in C_k$ and executes volume $v \le \min(q_{j,i}, q_{j,i'})$ on pair (i, i') whenever the crossed spread exceeds $c_{j,k} > 0$. All bots run identical standardised arbitrage software, so $\kappa_j = \kappa > 0$ for all j; this common *execution rate constant* is a known system parameter, which ensures that total deployed capital $\sum_j q_{j,i}$ is the sole free variable determining oracle security.

Define the *latency efficiency function* $g : [0, \infty)^2 \to [0, 1]$ is the probability that an arbitrage opportunity on pair (i, i') has not been captured by other market participants before the bot can observe and act on it. Modelling opportunity lifetimes as $\operatorname{Exp}(1/\tau_{\max})$, this probability is $e^{-(\ell+\ell')/\tau_{\max}}$, approximated to first order as the canonical linear choice

$$g(\ell, \ell') \;:=\; \left(1 - \frac{\ell + \ell'}{\tau_{\max}}\right)^+$$

The *effective arbitrage rate for pair (i, i') under placement S* is

$$\gamma_{k,ii'}(S) \;:=\; \kappa \sum_{j=1}^{N_k} \min(q_{j,i}, q_{j,i'}) \cdot g\big(\lambda(i, S), \lambda(i', S)\big) \tag{4}$$

and the *cluster effective arbitrage rate* is

$$\gamma_k(S) \;:=\; \min_{(i,i') \in C_k^2} \gamma_{k,ii'}(S) \tag{5}$$

the bottleneck pair rate governing the overall convergence speed.

Remark 2.5. The function $\gamma_k(S)$ is jointly determined by capital deployment and server placement via $g(\lambda(i,S), \lambda(i',S))$. Bots observing both exchanges at zero latency ($\lambda = 0$) achieve maximum rate $\kappa_j \min(q_{j,i}, q_{j,i'})$. As placement worsens (larger λ), efficiency degrades continuously to zero at $\lambda(i,S) + \lambda(i',S) = \tau_{\max}$. This coupling makes the convergence and placement problems inseparable, which motivates the unified treatment of this section.

2.3 Per-Exchange Order Flow and Convergence

Order flow is heterogeneous across exchanges within a cluster: a liquid exchange like Binance generates far more events per second than a thin venue.

Assumption 2.6. In cluster C_k, exchange $i \in C_k$ receives exogenous order flow as an independent Poisson process of intensity $\lambda_i^{(k)} > 0$. Each event on exchange i increments the crossed spread Δ_k by at most η_i (one tick on exchange i). The *cluster-level flow rate* is

$$\Lambda_k \;:=\; \sum_{i \in C_k} \lambda_i^{(k)} \eta_i \tag{6}$$

the total spread-inflation rate in dollars per unit time.

Remark 2.7. The cluster rate Λ_k in (6) weights each exchange's contribution by its tick size η_i: a liquid exchange with small tick but high frequency contributes differently from a thin exchange with large tick and low frequency.

Theorem 2.8. *Under Assumption 2.21 and placement S, starting from $\Delta_k(0) = \delta_0 \geq 0$:*

$$\mathbb{E}[\Delta_k(t)] \;\leq\; \delta_0\, e^{-\gamma_k(S)\,t} + \frac{\Lambda_k}{\gamma_k(S)}\big(1 - e^{-\gamma_k(S)\,t}\big) \tag{7}$$

In the stationary regime:

$$\mathbb{E}[\Delta_k(\infty)] \;\leq\; \frac{\Lambda_k}{\gamma_k(S)} \tag{8}$$

The system reaches $\mathcal{K}_\varepsilon$ in expectation if and only if

$$\gamma_k(S) \;\geq\; \frac{\Lambda_k}{\varepsilon} \tag{9}$$

Proof. The dynamics of $\mathbb{E}[\Delta_k(t)]$ are driven by two forces. The *arbitrage force*: whenever $\Delta_k > c_{j,k}$, bot j closes the spread at rate $\kappa_j \cdot \Delta_k \cdot g(\lambda(i,S), \lambda(i',S))$ for the relevant pair (i,i'); by Definition 2.4, the aggregate rate is $\geq \gamma_k(S) \cdot \Delta_k$. The *flow force*: the superposition of independent Poisson processes on all $i \in C_k$ contributes spread at mean rate $\Lambda_k = \sum_i \lambda_i^{(k)} \eta_i$ (by the superposition theorem for Poisson processes and linearity of expectation). Combining:

$$\frac{d}{dt}\mathbb{E}[\Delta_k(t)] \;\leq\; -\gamma_k(S)\,\mathbb{E}[\Delta_k(t)] + \Lambda_k$$

This linear ODE $\dot{y} \leq -\gamma_k(S)\, y + \Lambda_k$ has solution (15). The stationary bound (8) follows by $t \to \infty$; the equivalence (9) is immediate from $\Lambda_k/\gamma_k(S) \leq \varepsilon$. □

Corollary 2.9. *Fix a target $\mathbb{E}[\Delta_k(\infty)] \leq \varepsilon$. By (9), any combination of capital and placement achieving $\gamma_k(S) \geq \Lambda_k/\varepsilon$ is sufficient. Increasing capital (larger $q_{j,i}$) and improving placement (smaller $\lambda(i,S)$, hence larger g) are substitutes along the iso-performance curve $\gamma_k(S) = \Lambda_k/\varepsilon$. The marginal rate of substitution is:*

$$\left.\frac{\partial q_{j,i}}{\partial \lambda(i,S)}\right|_{\gamma_k = const} = -\frac{\kappa_j \min(q_{j,i}, q_{j,i'})}{(\Lambda_k/\varepsilon)} \cdot \frac{\partial g/\partial \lambda(i,S)}{g(\lambda(i,S), \lambda(i',S))}$$

For the canonical linear g: $\partial g/\partial \lambda = -1/\tau_{\max}$, so $\frac{\partial q}{\partial \lambda}\big| = -\kappa_j q/(\tau_{\max} \cdot \Lambda_k/\varepsilon)$. Improved placement (1 ms less latency) substitutes for $\kappa_j q/(1000\tau_{\max}^{-1} \cdot \Lambda_k/\varepsilon)$ dollars of additional capital.

2.4 Information Percolation and the Engine

Convergence measures how fast bots close intra-cluster spreads; percolation measures how fast a new price signal at any exchange reaches the Engine. Both depend on S.

Definition 2.10. The *percolation time* $T_{\text{perc}}(i, S)$ is the elapsed time from a price update at exchange i to the Engine broadcasting the revised f_t. It decomposes as

$$T_{\text{perc}}(i, S) \;=\; \underbrace{\lambda(i, S)}_{\text{obs. latency}} \;+\; \underbrace{T_{\text{conv},k(i)}}_{\text{intra-cluster conv.}} \;+\underbrace{\ell(\text{agg}_{k(i)}, e^*)}_{\text{agg. to Engine}}+\; \underbrace{\delta_{\text{proc}}}_{\text{COB compute}} \tag{10}$$

where $k(i)$ is exchange i's cluster, e^* is the Engine location, and $T_{\text{conv},k(i)}$ is the random time for $\Delta_{k(i)}$ to reach $\mathcal{K}_\varepsilon$ starting from the new deviation caused by the update.

Proposition 2.11. *Under placement S and Assumption 2.21:*

$$\mathbb{E}[T_{\text{perc}}(i, S)] \;\le\; \lambda(i, S) + \frac{1}{\gamma_{k(i)}(S)} \log \frac{\Delta p}{\varepsilon} + \ell(\text{agg}_{k(i)}, e^*) + \delta_{\text{proc}} \tag{11}$$

where Δp is the magnitude of the price update at exchange i.

Proof. The observation latency contributes $\lambda(i, S)$ deterministically. The intra-cluster convergence time: starting from $\Delta_{k(i)} = \Delta p$ after the update, Theorem 2.22 gives $\mathbb{E}[\Delta_{k(i)}(t)] \le \Delta p\, e^{-\gamma_{k(i)}(S)t} + \Lambda_{k(i)}/\gamma_{k(i)}(S)$. To reach $\mathbb{E}[\Delta_{k(i)}(t)] \le \varepsilon$, we need $\Delta p\, e^{-\gamma_{k(i)}(S)t} \le \varepsilon$, i.e. $t \ge \log(\Delta p/\varepsilon)/\gamma_{k(i)}(S) =: T_{\text{conv}}$. The aggregator-to-Engine and processing terms are deterministic. □

Definition 2.12. The *price-discovery weight* of exchange i at time t is $w_i^{(t)} := P_i^{(t)} \lambda_i^{(k)}$ where $P_i^{(t)} := d_i^{(t)} / \sum_{i'} d_{i'}^{(t)}$ is the depth share (see above). The stationary pair weight is $\bar{w}_{ii'} := \sum_r \pi_r^* w_{ii'}^{(r)}$ under the liquidity regime CTMC (Assumption 2.14). The *weighted percolation cost* of placement S is

$$\bar{\Psi}(S) \;:=\; \sum_{i \ne i'} \bar{w}_{ii'} \cdot \Big(\lambda(i, S) + \lambda(i', S) + \frac{1}{\gamma_k(S)} \log \frac{\bar{\Delta}}{\varepsilon} \Big) \tag{12}$$

where $\bar{\Delta}$ is the mean update magnitude. The first two terms capture observation latency; the third captures convergence time, both of which depend on S through $\lambda(\cdot, S)$ and $\gamma_k(S)$.

Remark 2.13. The objective $\bar{\Psi}(S)$ simultaneously penalises: (i) slow observation ($\lambda(i, S) + \lambda(i', S)$ large), and (ii) slow convergence ($\gamma_k(S)$ small, which happens when placement is poor and g is small). Both costs are minimised by placing servers near the highest-weight exchange pairs. This unification means the p-median placement (which minimises $\sum_{ii'} \bar{w}_{ii'} \min_f (\ell(i, f) + \ell(i', f))$, see Theorem 2.16) also approximately minimises the convergence time, since $\gamma_k(S)$ is increasing in g which is decreasing in $\lambda(i, S)$.

2.5 Placement as p-Median

Assumption 2.14 (Liquidity regime CTMC)**.** The liquidity state $\mathbf{R}(t) \in \mathcal{S} := \prod_i \mathcal{S}_i$ is an irreducible positive-recurrent CTMC with generator $\mathbf{Q} \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ and stationary distribution $\boldsymbol{\pi}^*$. In regime r, exchange i has depth share $P_i^{(r)}$ and order-flow rate $\lambda_i^{(r)}$, giving price-discovery weight $w_i^{(r)} := P_i^{(r)} \lambda_i^{(r)}$. The stationary weight $\bar{w}_i := \sum_r \pi_r^* w_i^{(r)}$ satisfies $T^{-1} \int_0^T w_i^{(t)}\, dt \xrightarrow{a.s.} \bar{w}_i$ as $T \to \infty$ by the CTMC ergodic theorem [7].

Definition 2.15. With stationary pair weights $\bar{w}_{ii'} := \sum_r \pi_r^* w_i^{(r)} w_{i'}^{(r)} / \sum_{j \neq j'} w_j^{(r)} w_{j'}^{(r)}$, the *stationary weighted closure cost* is

$$\bar{\Psi}_0(S) \;:=\; \sum_{i \neq i'} \bar{w}_{ii'} \cdot \min_{f \in S} \big(\ell(i,f) + \ell(i',f)\big) \tag{13}$$

This is the dominant term in $\bar{\Psi}(S)$ when $\gamma_k(S)$ is large; minimising $\bar{\Psi}_0$ approximates minimising $\bar{\Psi}$.

Theorem 2.16.

$$S^* := \arg\min_{|S|=N_s} \bar{\Psi}_0(S)$$

is the p-median problem on $(\mathcal{E} \times \mathcal{E}, \mathcal{F}, \ell, \bar{\mathbf{w}})$ *with* $p = N_s$.

Proof. $\bar{\Psi}_0(S) = \sum_r \pi_r^* \sum_{ii'} w_{ii'}^{(r)} \min_f(\ell(i,f) + \ell(i',f)) = \sum_{ii'} \bar{w}_{ii'} \min_f(\ell(i,f) + \ell(i',f))$ by exchanging finite sums. □

Theorem 2.17 (Complexity and approximation)**.** (i) *NP-hard [3].*

(ii) *Arya et al. [4] gives* $\bar{\Psi}_0(\hat{S}) \leq (1 + 1/e + \zeta)\bar{\Psi}_0(S^*)$ *in time polynomial in* $|\mathcal{E}|, |\mathcal{F}|, 1/\zeta$ *for fixed* N_s.

(iii) *LP gap* $\leq 1 + 1/e$ *[5]; gap* $= 1$ *on tree metrics.*

Theorem 2.18. *With spectral gap* $\zeta^* > 0$ *of* $\mathbf{Q}$ *and* $\Psi_{\max} := \max_{S,r} \Psi(S,r)$*:*

$$\frac{1}{T}\mathbb{E}\Big[\int_0^T \mathrm{Gap}(S^*, \mathbf{R}(t))\, dt\Big] \;\leq\; \frac{\Psi_{\max} \|\boldsymbol{\pi}^*(r_0) - \boldsymbol{\pi}^*\|_{\mathrm{TV}}}{\zeta^* T}\big(1 - e^{-\zeta^* T}\big) \;\xrightarrow{T \to \infty}\; 0$$

where $\mathrm{Gap}(S^*, r) := \Psi_0(S^*, r) - \min_{|S|=N_s} \Psi_0(S,r)$ *and* $\Psi_0(S,r) := \sum_{ii'} w_{ii'}^{(r)} \min_{f \in S}(\ell(i,f) + \ell(i',f))$.

Proof. $\|\mu_t - \boldsymbol{\pi}^*\|_{\mathrm{TV}} \leq \|\boldsymbol{\pi}^*(r_0) - \boldsymbol{\pi}^*\|_{\mathrm{TV}}\, e^{-\zeta^* t}$ by Perron–Frobenius; integrate over $[0,T]$; divide by T. □

Corollary 2.19 (Mixing time)**.** *The static placement* S^* *achieves time-average suboptimality* $\leq \delta_{\mathrm{gap}}$ *for all* $T \geq \Psi_{\max} \|\cdot\|_{\mathrm{TV}} / (\zeta^* \delta_{\mathrm{gap}})$*. For liquid markets (large* ζ^**), this mixing time is of order seconds.*

Theorem 2.20 (Optimal dynamic placement as MDP)**.** *With per-server switching cost* $c_{\mathrm{sw}} \geq 0$ *per regime change, the optimal policy* $\mathcal{P}^* : r \mapsto S_r$ *satisfies the Bellman equation*

$$V^*(r) = \min_{|S|=N_s}\Big[\Psi_0(S,r) + c_{\mathrm{sw}}|S \triangle S_{\mathrm{cur}}| + \frac{1}{\nu}\sum_{r'} Q_{rr'} V^*(r')\Big] \tag{14}$$

with $\nu := \max_r |Q_{rr}|$*. The policy satisfies:*

(i) $c_{\mathrm{sw}} = 0$*: always switch to* $\arg\min_S \Psi_0(S,r)$.

(ii) $c_{\mathrm{sw}} \to \infty$*: never switch (static* S^**).*

(iii) *Finite* c_{sw}*: switch from* S *to* S' *only when* $\Psi_0(S,r) - \Psi_0(S',r) > c_{\mathrm{sw}}|S' \triangle S|$.

(iv) *Solvable in polynomial time in* $|\mathcal{S}|$ *for fixed* N_s*,* $|\mathcal{F}|$ *via value iteration.*

Proof. Uniformise $\mathbf{Q}$ at rate ν; the average-cost MDP with reward $-\Psi_0(S,r) - c_{\mathrm{sw}}|S \triangle S_{\mathrm{prev}}|$ has optimality equation (14). Cases (i)–(ii) follow by inspection; (iii) is the standard switching-cost threshold [8]; (iv) holds for finite state and action spaces. □

2.6 Convergence Theorem

Assumption 2.21. Exogenous order flow in cluster C_k arrives as a Poisson process of intensity $\Lambda_k > 0$; each event increments Δ_k by at most η.

Theorem 2.22. *Under Assumption 2.21, starting from $\Delta_k(0) = \delta_0$:*

$$\mathbb{E}[\Delta_k(t)] \;\leq\; \delta_0\, e^{-\gamma_k t} + \frac{\Lambda_k \eta}{\gamma_k}(1 - e^{-\gamma_k t}) \tag{15}$$

In the stationary regime $\mathbb{E}[\Delta_k(\infty)] \leq \Lambda_k \eta / \gamma_k$. The system reaches $\mathcal{K}_\varepsilon$ in expectation if and only if $\gamma_k \geq \Lambda_k \eta / \varepsilon$.

Proof. The dynamics of $\mathbb{E}[\Delta_k(t)]$ satisfy $\frac{d}{dt}\mathbb{E}[\Delta_k] \leq -\gamma_k \mathbb{E}[\Delta_k] + \Lambda_k \eta$: bots reduce the spread at rate $\gamma_k \cdot \Delta_k$ (linear in the deviation, since execution rate is proportional to spread above $c_{j,k}$), while exogenous flow increases it at mean rate $\Lambda_k \eta$. Solving the linear ODE $\dot{y} \leq -\gamma_k y + \Lambda_k \eta$ gives (15). The stationary bound follows by $t \to \infty$, and the equivalence is direct from $\Lambda_k \eta / \gamma_k \leq \varepsilon$. □

3 Exchange Feed Manipulation and Detection

3.1 Feed Manipulation Model

Definition 3.1. Exchange i is said to deploy *feed manipulation* $\mathcal{M}_i$ if it presents to some subset of bots a snapshot $\tilde{x}_i^{(t)} \neq x_i^{(t,*)}$, where $x_i^{(t,*)}$ is the actionable ground truth. We parametrise the manipulation by its ℓ^1-*deviation*

$$\varepsilon_i^{(t)}(\mathcal{M}_i) := \left\| \tilde{x}_i^{(t)} - x_i^{(t,*)} \right\|_1$$

and its target $\mathcal{T}_i \subseteq [N_k]$: the bots that recieve the falsified feed. The manipulation is *selective* if $\mathcal{T}_i \subsetneq [N_k]$ and universal if $\mathcal{T}_i = [N_k]$. There are two canonical forms: (i) *IP-partitioning*: distinct snapshots served by IP subnet; (ii) *selective delay*: targeted bots receive updates stale by δ_{delay}, giving $\varepsilon_i^{(t)} \asymp \bar{\lambda}_i \mu_i \eta\, \delta_{\text{delay}}$.

Proposition 3.2. *Let agent j connect from a single IP $\alpha_j \in I_{\ell_j}$. No algorithm (computationally unbounded, deterministic or randomised) running solely on j's observations can distinguish an (ε, s)-IP-partitioned exchange from a legitimate exchange with ℓ^1-noise ε, with advantage exceeding zero.*

Proof. Agent j's view is the single sample $\tilde{x}^{(\ell_j)}$. Construct a coupling where the legitimate exchange sends $\tilde{x}^{(\ell_j)}$ (by setting its noise equal to $\tilde{x}^{(\ell_j)} - x^*$, of ℓ^1 norm $\leq \varepsilon$). The total variation distance between the two worlds as seen by j is therefore 0. □

3.2 Arbitrage-Driven Detection

Definition 3.3. Bot j receiving $\tilde{x}_i^{(t)}$ sees phantom cross-spread

$$\tilde{\Delta}_{j,ii'} := [b^{(t,*)}_{i',\beta(B^{C_k})} - a_{i,\alpha(\tilde{A}_i^{(t)})}]^+$$

A *phantom arbitrage* exists when $\tilde{\Delta}_{j,ii'} > c_{j,k}$ but $\Delta^{(t,*)}_{ii'} := [b^{(t,*)}_{i',\beta} - a_{i,\alpha(A_i^{(t,*)})}]^+ \leq c_{j,k}$. The *execution signal* is $\xi_j := p_{\text{fill}} - p_{\text{fake}}$, where $p_{\text{fake}} := a_{i,\alpha(\tilde{A}_i^{(t)})}$ and p_{fill} is the actual fill price.

Theorem 3.4. *Let exchange i deploy $\mathcal{M}_i$ with $\varepsilon_i^{(t)} > \varepsilon_{\min} := 2c_{j,k} + \sigma_{\text{exec}}$.*

(i) At bot j's next execution attempt, $\xi_j \geq (\varepsilon_i^{(t)} - \varepsilon_{\min})/2 > 0$ with probability 1.

(ii) Bot j detects the manipulation from its own execution alone; the rule flag i if $|\xi_j| > \sigma_{\text{exec}}$ on two consecutive attempts *has false-positive probability $< 1/4$.*

(iii) One flagging bot suffices to quarantine exchange i via Protocol $\mathcal{P}$ threshold signing.

(iv) Even when $\mathcal{T}_i = [N_k]$, every bot trading against exchange i receives signal $\xi_j > 0$.

Proof. (i). The falsified feed shows best ask $p_{\text{fake}} < a_{i,\alpha(A_i^{(t,*)})}$. The missing depth at and above p_{fake} is at least $\varepsilon_i^{(t)}/2$ (ask-side component of the ℓ^1 deviation). By Theorem 1.8, lifting v units when only $v - \varepsilon_i^{(t)}/2$ exist causes book-walk slippage of at least $\varepsilon_i^{(t)}/2$ per unit. Hence $\xi_j \geq \varepsilon_i^{(t)}/2 - c_{j,k} - \sigma_{\text{exec}} > 0$.

(ii). Bot observes $(p_{\text{fake}}, p_{\text{fill}})$ directly; gap $\xi_j > 0$ is observable. Two independent false positives have probability $(\sigma_{\text{exec}}/\xi_j)^2 < 1/4$.

(iii). Flag transmitted via the threshold-signed bot-to-aggregator channel.

(iv). Each bot trades against $x_i^{(t,*)}$ regardless; slippage $\xi_j > 0$ arises from the mismatch between falsified feed and actionable book. □

Corollary 3.5. *A selective delay δ_{delay} is detectable whenever*

$$\delta_{\text{delay}} > \frac{2c_{j,k} + \sigma_{\text{exec}}}{\bar{\lambda}_i \mu_i \eta}$$

In liquid markets this threshold is of order milliseconds.

3.3 Cluster Consistency Detection

Definition 3.6. $[\Sigma_i^{(t)}]_{jj'} := \tilde{\Delta}_{j,ii'}^{(t)} - \tilde{\Delta}_{j',ii'}^{(t)}$. Exchange i is cluster-consistent if $\left\|\Sigma_i^{(t)}\right\|_\infty \leq \tau_{\text{cons}}$.

Theorem 3.7. *(i) If $|\mathcal{T}_i| < N_k$: $\left\|\Sigma_i^{(t)}\right\|_\infty \geq \varepsilon_i^{(t)}/2$, so exchange i is flagged iff $\varepsilon_i^{(t)} > 2\tau_{\text{cons}}$.*

(ii) If $\mathcal{T}_i = [N_k]$: $\Sigma_i^{(t)} = 0$; detection falls to Theorem 3.4(iv).

(iii) For selective delay $\delta_{\text{delay}} > \tau_{\max}$, define $\tau_i^{(t)} := \max_{j \in \mathcal{T}_i, j' \notin \mathcal{T}_i} |\hat{t}_j^{\text{arb}} - \hat{t}_{j'}^{\text{arb}}|$. Then $\tau_i^{(t)} \geq \delta_{\text{delay}}$ with probability 1.

Proof. (i). For honest $j^* \notin \mathcal{T}_i$ and targeted $j \in \mathcal{T}_i$: $[\Sigma]_{j,j^*} = a_{i,\alpha(A_i^{(t,*)})} - a_{i,\alpha(\tilde{A}_i^{(t)})} \geq \varepsilon_i^{(t)}/2$ since the ask-side ℓ^1 deviation is $\geq \varepsilon_i^{(t)}/2$.

(ii). All $\tilde{\Delta}_{j,ii'}^{(t)}$ identical; $\Sigma_i^{(t)} = 0$.

(iii). Untargeted bots react at t^*; targeted bots at $t^* + \delta_{\text{delay}}$. Gap $\delta_{\text{delay}} > \tau_{\max}$ exceeds legitimate intra-cluster timing noise. □

3.4 Passive Oracle Impossibility

Definition 3.8. An oracle is *passive* if its reporters do not submit orders to any exchange in $\mathcal{E}$. Examples: Chainlink [9], Pyth [10], TWAP aggregators.

Theorem 3.9. *For any passive oracle $\mathcal{O}$ and any feed manipulation $\mathcal{M}_i$ with deviation $\leq \varepsilon$, there exists a legitimate feed with ℓ^1-noise $\leq \varepsilon$ indistinguishable from $\mathcal{M}_i$ by $\mathcal{O}$.*

Proof. A passive oracle's observation is the single snapshot $\tilde{x}_i^{(t)}$. By Proposition 3.2, the total variation distance between a manipulated feed and a legitimately noisy feed, as seen by any single-IP reporter, is 0. Since the oracle makes no trades, it has no execution signal and cannot consult additional ground truth. □

4 Manipulation Security and Optimal Delay

4.1 Attack Formulation

Definition 4.1. The (t, ε)-manipulation game: attacker $\mathcal{A}$ places and cancels orders subject to capital $W > 0$; oracle $\mathcal{O}$ publishes f_t after delay Δt. Attacker payoff:

$$\Pi(\mathcal{A}) := \varepsilon \cdot \mathrm{TVS} - \mathrm{Cost}(\mathcal{A})$$

Furthermore we define the depth and cost function respectively as

$$D^{(t)}(\varepsilon) := \sum_{i \in \mathcal{E}} \sum_{k: p_k \in [f_t - \varepsilon, f_t]} b_{i,k}^{(t)}$$
$$C^{(t)}(\varepsilon) := \sum_{i \in \mathcal{E}} \sum_{k: p_k \in [f_t - \varepsilon, f_t]} (f_t - p_k) \cdot b_{i,k}^{(t)}$$

Assumption 4.2. $\mathrm{diam}(C_k) \leq \tau_{\max} \ll \Delta t$ for all k; DAN convergence completes before oracle publication.

Lemma 4.3. *$D^{(t)}$ and $C^{(t)}$ are non-decreasing step functions with $C^{(t)}(\varepsilon) \leq \varepsilon D^{(t)}(\varepsilon)$ and $C^{(t)}(\varepsilon)/\varepsilon$ non-decreasing.*

Proof. Monotonicity: both sums grow weakly in ε. Bound: $f_t - p_k \leq \varepsilon$ for $p_k \in [f_t - \varepsilon, f_t]$. Non-decreasing ratio: additional levels at $\varepsilon' > \varepsilon$ cost $\geq \varepsilon$ per unit. □

4.2 Manipulation Cost Lower Bound

Theorem 4.4. *Under Assumption 4.2, any strategy skewing f_t downward by $\varepsilon > 0$ for duration Δt requires capital:*

$$W^*(\varepsilon, \Delta t) \;\geq\; C^{(t)}(\varepsilon) + D^{(t)}(\varepsilon) \cdot \varepsilon + \Lambda \cdot \Delta t \cdot \varepsilon \tag{16}$$

Proof. Three necessary components.

(i) Sweep cost $C^{(t)}(\varepsilon)$. All bids in $[f_t - \varepsilon, f_t]$ must be lifted; the net outlay above the $f_t - \varepsilon$ baseline is $\sum_{i,k}(f_t - p_k) \cdot b_{i,k}^{(t)} = C^{(t)}(\varepsilon)$.

(ii) Wall depth $D^{(t)}(\varepsilon)$ at cost $\varepsilon \cdot D^{(t)}(\varepsilon)$. After sweeping, the attacker must post ask depth at $f_t - \varepsilon$ to ensure the COB computes $\alpha(\tilde{A}) = f_t - \varepsilon$. The required depth is $D^{(t)}(\varepsilon)$ for the following reason: after sweeping all bids above $f_t - \varepsilon$, the residual best bid is below $f_t - \varepsilon$. For the COB to report fair price $f_t - \varepsilon/2$ (approximately), the best ask must equal $f_t - \varepsilon$, and this ask must not be consumed by the exhaustion loop. This requires the ask wall to dominate all residual asks at $f_t - \varepsilon$, meaning depth $\geq D^{(t)}(\varepsilon)$. The capital locked in this wall (at risk of being sniped at price ε below fair value) is $D^{(t)}(\varepsilon) \cdot \varepsilon$.

(iii) Maintenance $\Lambda \cdot \Delta t \cdot \varepsilon$. Exogenous market orders arrive at rate Λ during Δt, consuming wall depth at $f_t - \varepsilon$ at cost ε per unit. □

The bound (16) implicitly assumes the attacker maintains the sell wall for the full duration Δt. We now formalise the *reserve-constrained attack* in which the attacker cancels early, exploiting the fact that DAN bots cannot replenish capital instantaneously.

Definition 4.5. Let $\tau_{\mathrm{rep}} > 0$ be the *bot replenishment time*: the time required for a bot that has exhausted its capital on exchange i to transfer funds and resume sniping. Define $\tau_{\mathrm{cancel}} \in (0, \Delta t)$ as the time at which the attacker cancels their sell wall. The attacker's strategy is:

(i) Sweep bids in $[f_t - \varepsilon, f_t]$ at cost $C^{(t)}(\varepsilon)$.

(ii) Post sell wall of depth $D^{(t)}(\varepsilon)$ at $f_t - \varepsilon$.

(iii) Wait τ_{cancel} seconds.

(iv) Cancel the remaining wall.

In $[0, \tau_{\text{cancel}}]$, DAN bots snipe the wall until either the wall is exhausted or their capital runs out. Let $Q_k^{(\tau_{\text{cancel}})}$ denote the capital deployable within time τ_{cancel}.

Lemma 4.6. *If bots can deploy their full capacity $Q_k^{(\Delta t)}$ over Δt at uniform rate, then the capital deployable within τ_{cancel} is*

$$Q_k^{(\tau_{\text{cancel}})} \;=\; Q_k^{(\Delta t)} \cdot \min\left(1,\, \frac{\tau_{\text{cancel}}}{\tau_{\text{rep}}}\right)$$

For $\tau_{\text{cancel}} < \tau_{\text{rep}}$, bots can only deploy a fraction of their total capital before the attacker cancels.

Proof. Bot capital depletes at rate $Q_k^{(\Delta t)}/\tau_{\text{rep}}$ per unit time (the rate at which capital is consumed in arbitrage). Over τ_{cancel} seconds, total deployable capital is $Q_k^{(\Delta t)} \cdot \tau_{\text{cancel}}/\tau_{\text{rep}}$, capped at $Q_k^{(\Delta t)}$. □

Theorem 4.7. *Under the reserve-constrained attack with cancellation time $\tau_{\text{cancel}} < \tau_{\text{rep}}$, the attacker's net cost is*

$$W_{\text{rc}}(\varepsilon, \tau_{\text{cancel}}) \;=\; C^{(t)}(\varepsilon) + \varepsilon \cdot Q_k^{(\Delta t)} \cdot \frac{\tau_{\text{cancel}}}{\tau_{\text{rep}}} + \Lambda \cdot \tau_{\text{cancel}} \cdot \varepsilon \tag{17}$$

and the oracle price f_t is skewed by ε for the epoch if $\tau_{\text{cancel}} > \Delta t/2$ (the wall persists past the observation window used to compute f_t). The attack is profitable when

$$\varepsilon \cdot \text{TVS} \;>\; C^{(t)}(\varepsilon) + \varepsilon \cdot Q_k^{(\Delta t)} \cdot \frac{\tau_{\text{cancel}}}{\tau_{\text{rep}}} + \Lambda \cdot \tau_{\text{cancel}} \cdot \varepsilon \tag{18}$$

In particular, for $\tau_{\text{cancel}} \to 0$, the attack cost approaches $C^{(t)}(\varepsilon)$ — the sweep cost alone — which is much less than $W^*(\varepsilon, \Delta t)$ in (16).

Proof. The sweep cost $C^{(t)}(\varepsilon)$ is unavoidable. The DAN snipes $V_{\text{snipe}} = \min(D^{(t)}(\varepsilon), Q_k^{(\tau_{\text{cancel}})})$ volume during $[0, \tau_{\text{cancel}}]$; by Lemma 4.6, $Q_k^{(\tau_{\text{cancel}})} = Q_k^{(\Delta t)} \cdot \tau_{\text{cancel}}/\tau_{\text{rep}}$ for $\tau_{\text{cancel}} < \tau_{\text{rep}}$. Each sniped unit costs the attacker ε. The maintenance cost over τ_{cancel} is $\Lambda \tau_{\text{cancel}} \varepsilon$. After cancellation, the remaining wall depth $D^{(t)}(\varepsilon) - V_{\text{snipe}}$ is returned to the attacker (no further cost). Summing gives (17). □

Remark 4.8. Equation (17) reveals that the attacker holds an implicit option: they can observe whether the oracle is likely to update based on the manipulated book and cancel early if the manipulation is detected. The value of this option is $W^*(\varepsilon, \Delta t) - W_{\text{rc}}(\varepsilon, \tau_{\text{cancel}})$, which is strictly positive for $\tau_{\text{cancel}} < \tau_{\text{rep}}$.

4.3 Optimal Publication Delay

The publication delay Δt plays two opposing roles. Increasing Δt raises the manipulation cost via the maintenance term $\Lambda \Delta t \varepsilon$ in (16) and also forces $\tau_{\text{cancel}} \geq \Delta t$ (making early cancellation impossible if $\Delta t > \tau_{\text{rep}}$). However, a larger Δt also makes the oracle stale: during Δt, the true price moves by approximately $\sigma\sqrt{\Delta t}$ (diffusion model), creating a front-running opportunity worth $\sigma\sqrt{\Delta t} \cdot \text{TVS}$ for anyone who can predict near-term price direction.

Definition 4.9. Under a diffusion model for the true price with volatility $\sigma > 0$, the expected value of the front-running option created by staleness Δt is

$$F(\Delta t) \;:=\; \frac{1}{2}\sigma\sqrt{\Delta t} \cdot \text{TVS}$$

Theorem 4.10. *Define the* oracle vulnerability *under delay Δt and manipulation target ε as*

$$V(\Delta t, \varepsilon) \;:=\; F(\Delta t) + \left[\varepsilon \cdot \text{TVS} - W^*(\varepsilon, \Delta t)\right]^+ \tag{19}$$

where $W^*(\varepsilon, \Delta t)$ is the full manipulation cost (16). The optimal publication delay is

$$\Delta t^* \;:=\; \underset{\Delta t > \tau_{\text{rep}}}{\arg\min}\, V(\Delta t, \varepsilon) \tag{20}$$

where the constraint $\Delta t > \tau_{\text{rep}}$ eliminates the reserve-constrained attack. Under the linear depth model $D^{(t)}(\varepsilon) = \rho\varepsilon$ and the approximation $C^{(t)}(\varepsilon) \approx \rho\varepsilon^2/2$, the interior minimiser satisfies

$$\Delta t^* \;=\; \max\left(\tau_{\text{rep}},\; \frac{\sigma^2 \text{TVS}^2}{4\Lambda^2\varepsilon^2}\right) \tag{21}$$

Proof. The manipulation component $[\varepsilon \cdot \text{TVS} - W^*(\varepsilon, \Delta t)]^+$ is non-increasing in Δt: increasing Δt increases the maintenance cost $\Lambda \Delta t \varepsilon$, reducing attack profitability. For large enough Δt, manipulation is unprofitable and this term is zero. The staleness component $F(\Delta t) = \frac{1}{2}\sigma\sqrt{\Delta t} \cdot \text{TVS}$ is strictly increasing in Δt. The sum has an interior minimum (when the manipulability is non-zero) or is minimised at the smallest Δt that makes manipulation unprofitable.

Under the linear depth model, $W^*(\varepsilon, \Delta t) = \rho\varepsilon^2/2 + \rho\varepsilon^2 + \Lambda\Delta t\varepsilon$. Manipulation is profitable when $\varepsilon \cdot \text{TVS} > W^*$, i.e. $\Lambda\Delta t\varepsilon < \text{TVS}\varepsilon - (3\rho/2)\varepsilon^2$, i.e. $\Delta t < (\text{TVS} - (3\rho/2)\varepsilon)/\Lambda$. In this regime, $V(\Delta t, \varepsilon) = \frac{1}{2}\sigma\sqrt{\Delta t} \cdot \text{TVS} + \varepsilon \cdot \text{TVS} - W^*(\varepsilon, \Delta t)$, and $\partial V/\partial(\Delta t) = \sigma\text{TVS}/(4\sqrt{\Delta t}) - \Lambda\varepsilon = 0$ gives $\sqrt{\Delta t^*} = \sigma\text{TVS}/(4\Lambda\varepsilon)$, i.e. $\Delta t^* = \sigma^2\text{TVS}^2/(16\Lambda^2\varepsilon^2)$. Taking the maximum with τ_{rep} enforces the hard constraint that eliminates reserve-constrained attacks. □

Corollary 4.11. *The optimal delay Δt^* is:*

- Increasing in σ *(higher volatility → longer delay needed to make maintenance cost dominate staleness cost).*
- Decreasing in Λ *(higher order-flow rate → maintenance cost per unit time is higher, so shorter delay suffices to deter manipulation).*
- Decreasing in ε *(smaller target deviation → attack is already expensive, so delay can be shorter).*
- Lower bounded by τ_{rep} *(hard constraint from the reserve-constrained attack).*

4.4 DAN Capital as Counterforce

Definition 4.12. Cluster C_k is *Q-secure* if

$$Q_k^{(\Delta t)} := \sum_j \min\Big(q_{j,i}, \sum_{i'} q_{j,i'}\Big) \geq Q$$

Lemma 4.13. *If C_k is Q-secure and $\Delta t \geq \tau_{\text{rep}}$, the DAN snipes $V_{\text{snipe}} = \min(D^{(t)}(\varepsilon), Q)$ wall volume within Δt, costing the attacker $\varepsilon \cdot V_{\text{snipe}}$.*

Proof. When $\Delta t \geq \tau_{\text{rep}}$, bots can replenish capital within the epoch. Total deployed capital reaches $Q_k^{(\Delta t)} \geq Q$. Each bot buys up to $\min(q_{j,i}, \sum_{i'} q_{j,i'})$ units at $f_t - \varepsilon$ and sells at $\approx f_t$, earning ε per unit and depleting the attacker's wall. □

4.5 Global Security Theorem

Theorem 4.14. *Under **(H1)** Protocol $\mathcal{P}$ with $t_k > N_k/2$; **(H2)** Q-security with $Q \geq D^{(t)}(\varepsilon)$; **(H3)** $\Delta t \geq \tau_{\text{rep}}$ and $\Delta t > \max_k \operatorname{diam}(C_k)$:*

$$\Pi(\mathcal{A}) \leq \varepsilon \cdot \text{TVS} - W^*(\varepsilon, \Delta t) - \varepsilon D^{(t)}(\varepsilon) \leq 0$$

whenever $\varepsilon \cdot \text{TVS} \leq W^*(\varepsilon, \Delta t) + \varepsilon D^{(t)}(\varepsilon)$.

Proof. By (H3), $\Delta t \geq \tau_{\text{rep}}$ eliminates the reserve-constrained attack (Theorem 4.7); by (H2) and Lemma 4.13, DAN snipes $D^{(t)}(\varepsilon)$ wall volume (additional attacker loss $\varepsilon D^{(t)}(\varepsilon)$); by Theorem 4.4 the sweep+wall cost is $W^*(\varepsilon, \Delta t)$. □

Corollary 4.15. *Under the linear depth model $D^{(t)}(\varepsilon) \geq \rho\varepsilon$, all attacks above*

$$\varepsilon^* := \frac{\text{TVS}}{\frac{3\rho}{2} + \Lambda \Delta t}$$

are unprofitable. (Under the linear model, $W^*(\varepsilon, \Delta t) \geq C^{(t)}(\varepsilon) \geq \rho\varepsilon^2/2$ and $\varepsilon D^{(t)}(\varepsilon) \geq \rho\varepsilon^2$; profitability requires $\text{TVS} \geq \varepsilon(3\rho/2 + \Lambda\Delta t)$.)

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